

Supplementary Information for HIP: Hessian Interatomic Potentials without derivatives

Andreas Burger¹, Luca Thiede¹, Nikolaj Rønne¹,
Nandita Vijaykumar¹, Varinia Bernales¹, Tejs Vegge¹,
Arghya Bhowmik¹, Alán Aspuru-Guzik^{1*}

¹Affiliations are given in the main text..

1 Interatomic potentials and equivariant neural networks

Early works on machine learning interatomic potentials used local atomic environment descriptors in combination with linear regression [1, 2], Gaussian processes [3], and simple neural networks [4]. SchNet [5] introduced *rotation-invariant graph neural networks* for predicting molecular properties such as energies and forces. In parallel, *rotation-equivariant* architectures were developed, such as Cormorant [6], DimeNet [7], GemNet [8], PaiNN [9], NequIP [10], SphereNet [11], ViSNet [12], QuinNet [13], MACE [14] and the Equiformer family [15–17].

After several layers of message passing, task-dependent readout heads map the node features to the desired targets while respecting symmetry. For example, energies E are $\text{SO}(3)$ -invariant per-graph scalars. The energy readout head therefore uses a global pooling and a reduction operation to output an $l=0$ feature. Forces $\mathbf{F}$ are $\text{SO}(3)$ equivariant per-atom vectors ($l=1$), and can be output either by a direct force readout head or as the derivative of the energy $\mathbf{F} = -\nabla_{\mathbf{R}} E$. Hessians are $\text{SO}(3)$ -equivariant per-graph matrices composed of per-atom-pair 3×3 equivariant sub-matrices.

There is a rich connection between the HIP Hessian prediction in this work and other rotation-equivariant matrices of interest, like Hamiltonian prediction. For example, PhisNet [18], QHNet [19], HELM [20], and WANet [21] predicts the Fock matrix, and Graph2Mat [22] targets the density matrix. MōLe [23, 24] predicts coupled cluster amplitudes from Hartree-Fock orbitals.

2 Traditional Hessian methods

2.1 Analytical Hessians via CPKS

The gold standard for computing nuclear Hessians is to derive them analytically. In Kohn–Sham density functional theory (DFT), this is done with the Coupled Perturbation Kohn–Sham (CPKS) method. This is much more complex and computationally demanding than energy or force calculations. The difficulty arises from the need to calculate the response term of the electron density, which requires understanding how the self-consistent field (SCF) solution to the Kohn–Sham equations changes with changes in nuclear coordinates. To see why this is necessary, write the DFT energy in terms of the density matrix $\mathbf{P}(\mathbf{R}) = (\mathbf{C})^T(\mathbf{R})\mathbf{S}(\mathbf{R})\mathbf{C}(\mathbf{R})$ with molecular orbital coefficients $\mathbf{C}$ and overlap matrix $\mathbf{S}$:

$$E[\mathbf{P}] = \text{Tr}[h\mathbf{P}] + \frac{1}{2}\text{Tr}[\mathbf{J}[\mathbf{P}]] + \rho(\mathbf{P}) \quad (1)$$

To get the force, so the nuclear gradient, we define the Lagrangian to enforce normalization of the wavefunction:

$$\mathcal{L}(C, \epsilon, \mathbf{R}) = E[\mathbf{R}] - \text{Tr}[\epsilon(\mathbf{C}^T\mathbf{S}\mathbf{C} - \mathbf{I})] \quad (2)$$

We then get the nuclear gradient by differentiating $\mathcal{L}$. The derivative w.r.t. nuclear component x is:

$$\frac{d\mathcal{L}}{dx} = \frac{\partial\mathcal{L}}{\partial x} + \underbrace{\frac{\partial\mathcal{L}}{\partial\mathbf{C}}}_{=0} \frac{\partial\mathbf{C}}{\partial x} - \text{Tr}\left[\epsilon\mathbf{C}^T \frac{\partial\mathbf{S}}{\partial x} \mathbf{C}\right] + 2\text{Tr}\left[(\mathbf{S}\mathbf{C}\epsilon)^T \frac{d\mathbf{C}}{dx}\right] \quad (3)$$

$$= \frac{\partial E}{\partial x} - 2\text{Tr}\left[(\mathbf{S}\mathbf{C}\epsilon)^T \frac{d\mathbf{C}}{dx}\right] - \text{Tr}\left[\epsilon\mathbf{C}^T \frac{\partial\mathbf{S}}{\partial x} \mathbf{C}\right] + 2\text{Tr}\left[(\mathbf{S}\mathbf{C}\epsilon)^T \frac{d\mathbf{C}}{dx}\right] \quad (4)$$

$$= \frac{\partial E}{\partial x} - \underbrace{\text{Tr}\left[\epsilon\mathbf{C}^T \frac{\partial\mathbf{S}}{\partial x} \mathbf{C}\right]}_{\text{"Pulay terms"}} \quad (5)$$

Due to the stationarity $\frac{\partial\mathcal{L}}{\partial\mathbf{C}} = 0$ we did not need to calculate the density response $\frac{\partial\mathbf{C}}{\partial x}$, which is why calculating forces is cheap, on the same order as the cost of energies. However, once we want Hessians, we need to differentiate (2) twice:

$$\frac{d^2\mathcal{L}}{dxdy} = \frac{\partial^2\mathcal{L}}{\partial x\partial y} + \frac{\partial^2\mathcal{L}}{\partial\mathbf{C}\partial x} \frac{d\mathbf{C}}{dy} + \frac{\partial^2\mathcal{L}}{\partial\epsilon\partial x} \frac{d\epsilon}{dy} \quad (6)$$

$$= \frac{\partial^2\mathcal{L}}{\partial y\partial x} - \left[\frac{\partial^2\mathcal{L}}{\partial\mathbf{C}\partial x} \frac{\partial^2\mathcal{L}}{\partial\epsilon\partial x}\right] \left[\frac{\partial^2\mathcal{L}}{\partial\mathbf{C}^2} \frac{\partial^2\mathcal{L}}{\partial\mathbf{C}\partial\epsilon} \frac{\partial^2\mathcal{L}}{\partial\epsilon\partial\mathbf{C}} \frac{\partial^2\mathcal{L}}{\partial\epsilon^2}\right]^{-1} \left[\frac{\partial^2\mathcal{L}}{\partial\mathbf{C}\partial y} \frac{\partial^2\mathcal{L}}{\partial\epsilon\partial y}\right]. \quad (7)$$

Solving the linear system of equations scales formally as $O(N^5)$ in computation and $O(N^4)$ in memory. Similar to energy calculations, density fitting, sparsification (integral screening), and other optimizations can reduce the cost for CPKS. Still, the scaling remains worse in both computation and memory compared to energy calculations.

2.2 Hessians via finite differences

A major downside of the CPKS approach to Hessians is its highly complex implementation and need for exchange-functional-specific derivations. Therefore, many methods do not support Hessians, and most codes resort to finite-difference calculations in such cases. Using central differences of the forces along the $3N$ Cartesian directions, the Hessian elements are constructed from

$$\mathbf{H}_{ij} \approx \frac{\mathbf{F}_i(\mathbf{R} + h\mathbf{e}_j) - \mathbf{F}_i(\mathbf{R} - h\mathbf{e}_j)}{2h} \quad (8)$$

This requires $2 \times 3N$ displaced *gradient* calculations. Each gradient calculation scales similarly to the energy calculation with $O(N^4)$, leading to $O(N^5)$, the same as naive CPKS, although typically with more numerical noise and a much larger prefactor because we have to repeatedly converge an SCF, as opposed to solving a large linear system in analytical Hessians.

3 Hessians in Molecular Optimization

3.1 Geometry optimization

Geometry optimization locates minima $\text{argmin}_{\mathbf{R}} E(\mathbf{R})$ of the potential energy surface starting from non-equilibrium molecular geometries. It underpins most workflows from global searches[35, 36] to high-throughput screening[37, 38]. The efficiency of geometry optimization depends on four key factors: (i) the choice of coordinate system, (ii) the quality of the Hessian approximation, (iii) the update strategy, and (iv) the step size control. Cartesian coordinates are straightforward but strongly coupled, limiting step size and slowing convergence for flexible systems. In contrast, internal coordinates are better aligned with chemically meaningful motions, separate stiff and soft modes, and allow constraints to be imposed naturally. These properties often lead to substantially fewer iterations. Some of the most widely used optimizers are the first-order FIRE [39] optimizer and the second-order rational function optimization (RFO) optimizer.

3.1.1 RFO

Rational function optimization (RFO) [40, 41] is a commonly used second-order optimization technique in molecular geometry optimization. RFO starts with a $[2/2]$ -Padé-expansion of the energy:

$$E(\mathbf{R}_t + \Delta\mathbf{x}) - E(\mathbf{R}_t) \approx \frac{\mathbf{g}^\top \Delta\mathbf{x} + \frac{1}{2} \Delta\mathbf{x}^\top \mathbf{H} \Delta\mathbf{x}}{1 + |\Delta\mathbf{x}|^2} \quad (9)$$

The extrema $\Delta x'$ of this surrogate are given by the solution of the generalized eigenvalue problem:

$$\begin{bmatrix} \mathbf{H} & \mathbf{g} \\ \mathbf{g}^\top & 0 \end{bmatrix} \begin{bmatrix} \Delta \mathbf{x}' \\ 1 \end{bmatrix} = \lambda \begin{bmatrix} I & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \Delta \mathbf{x}' \\ 1 \end{bmatrix}. \quad (10)$$

An attractive property of RFO is that it can be used both for minimizing $E(\mathbf{R})$ and for finding saddle points. Selecting the lowest eigenvector of the augmented problem in Eq. (10) yields an update toward a minimum; selecting the second eigenvector yields a transition-state update. In contrast to Newton–Raphson optimization, RFO is also robust to indefinite Hessians. For a detailed derivation, see [40, 41]. For minimization we use restricted-step RFO (RS-RFO), which adds a trust region to prevent unphysically large steps. For transition-state search we use restricted-step partitioned RFO (RS-P-RFO), which treats the subspaces with negative and positive Hessian eigenvalues separately.

3.1.2 BFGS

To apply RFO in practice, we need a Hessian at each step. As computing full Hessians is usually too expensive, a common practice is to maintain an approximate Hessian using the BFGS quasi-Newton scheme. BFGS updates approximate Hessians $\mathbf{B}_k$ of the true Hessian $\mathbf{H}(\mathbf{R}_k)$ using the update equation

$$\mathbf{B}_{k+1} = \mathbf{B}_k - \frac{\mathbf{B}_k \mathbf{s}_k \mathbf{s}_k^\top \mathbf{B}_k}{\mathbf{s}_k^\top \mathbf{B}_k \mathbf{s}_k} + \frac{\mathbf{y}_k \mathbf{y}_k^\top}{\mathbf{y}_k^\top \mathbf{s}_k}, \quad \mathbf{y}_k^\top \mathbf{s}_k > 0, \quad (11)$$

where the curvature condition $\mathbf{y}_k^\top \mathbf{s}_k > 0$ (typically ensured by a Wolfe line search) guarantees that $\mathbf{B}_{k+1}$ remains positive-definite, which is desirable for minimization. In the RFO framework, $\mathbf{B}_k$ simply replaces the exact Hessian $\mathbf{H}$ in the augmented eigenvalue problem in (10).

3.2 Transition-state search

Transition states (TSs) are first-order saddle points on the potential energy surface (PES). They represent the maximum barrier along the minimum energy path (MEP) between two minima. Identifying transition states is crucial for understanding reaction mechanisms and mapping reaction networks. A first-order saddle point is characterized by having exactly one negative eigenvalue of the Hessian. Over the years, various computational methods have been developed to describe MEPs and locate transition states.

Computational methods for transition state search can broadly be categorized as *single-ended* and *double-ended*. In double-ended methods, we know the product and reactant states (the two minima on the energy surface) and seek an interpolation along the MEP that passes through the TS. In contrast, in single-ended methods, we only know a starting point and climb the energy surface to find a nearby transition state. Often, double-ended methods are used to give good initial guesses, which are then

refined by single-ended methods. In this study, we use the Growing String method to find the initial states, which we then refine using RS-P-RFO (see above).

Different double-ended methods are distinguished by how they grow the interpolations between product and reactant states. The most common approaches are the nudge elastic band method [42, 43] and the growing string method (GSM) [44, 45].

The nudged elastic band method aims to find the minimum energy path between given initial and final configurations. A set of images connecting these states is linked by classical spring forces, forming an elastic band. Each image experiences a total force comprising the spring force along the local tangent and the true force perpendicular to it. The images are simultaneously optimized to trace the MEP. To converge to the transition state, a climbing image is introduced, for which the force along the band is inverted, leaving only the perpendicular component of the true force. This drives the climbing image up the potential energy surface along the band while descending perpendicularly, reaching the transition state along the MEP.

The growing string method (GSM) locates transition states by incrementally constructing a discrete path of structures on the potential energy surface. Starting from the reactant and product endpoints, two path fragments are iteratively grown together. At each iteration, an additional node on the string is added along the local tangent direction and relaxed orthogonally to the path according to the force

$$\mathbf{F}_\perp = \mathbf{F} - (\hat{\mathbf{t}}^\top \mathbf{F}) \hat{\mathbf{t}}, \quad (12)$$

where $\mathbf{F} = -\nabla E$ is the force and $\hat{\mathbf{t}}$ is the local tangent unit vector. Convergence is typically accelerated by approximate Hessians, constructed in delocalized internal coordinates and updated by quasi-Newton schemes, which enable eigenvector-guided optimization while avoiding full second-derivative evaluations [45]. Once the two fragments merge, the full string is reparameterized to maintain uniform node distance, and the highest-energy node serves as a transition state estimate, which can be further refined using Hessian-based eigenvectors [44].

3.3 Frequency analysis for extrema classification

The final step in a transition state search or minima optimization is to confirm convergence via frequency analysis. A transition state is identified by the Hessian having exactly one negative eigenvalue (or equivalently, one imaginary frequency of the mass-weighted Hessian), whereas a minimum contains no imaginary frequencies. To carry out frequency analysis, one must remove the five or six redundant degrees of freedom, corresponding to the invariance of the energy under rotation and translation. This is done by mass-weighting the Hessian and performing an Eckart projection [46]. Then, the projected matrix is decomposed into its eigenvalues, with all positive eigenvalues indicating a minimum and exactly N negative eigenvalues indicating the presence of an order- N transition state.

3.4 Zero-point energy

Zero-point energy (ZPE) corrections account for the quantum mechanical vibrational energy that molecules have at absolute zero temperature. To calculate the

ZPE, one relaxes a geometry to an extremum, computes the Hessian, and sums the frequencies $ZPE = \frac{\hbar}{2} \sum_i \sqrt{\tilde{\lambda}_i}$. Here, $\tilde{\lambda}_i$ are eigenvalues of the mass-weighted Hessian $\tilde{H}_{ij} = H_{ij} / \sqrt{m_i m_j}$, where m_i and m_j are the atomic masses associated with Cartesian coordinates i and j , and $\hbar$ is Planck's reduced constant. One is usually interested in the relative ZPE between reactant and product states: $\Delta ZPE = ZPE(\text{product}) - ZPE(\text{reactant})$. The relative ZPE is relevant for reaction thermochemistry, as ΔZPE enters as a correction to the reaction free energy ΔG and therefore the equilibrium constant, which relates forward and backward rates via detailed balance.

4 Hessian properties

The Hessian is a symmetric matrix $\mathbf{H} = \mathbf{H}^\top$ where each sub-block transforms under rotation as a Cartesian tensor

$$\mathbf{H}_{I,J} \xrightarrow{\mathbf{Q}} \mathbf{Q} \mathbf{H}_{I,J} \mathbf{Q}^\top \quad (13)$$

or equivalently

$$\mathbf{H} \xrightarrow{\mathbf{Q}} (\mathbf{I}_N \otimes \mathbf{Q}) \mathbf{H} (\mathbf{I}_N \otimes \mathbf{Q})^\top \quad (14)$$

The symmetry follows directly from Schwarz's theorem, which states that for every scalar function $E(\mathbf{R})$ that has continuous partial second derivatives in the neighborhood of a point $\mathbf{R}_0$, the partial second derivatives commute

$$\left(\frac{\partial^2 E}{\partial \mathbf{R}_i \partial \mathbf{R}_j} \right)_{\mathbf{R}=\mathbf{R}_0} = \left(\frac{\partial^2 E}{\partial \mathbf{R}_j \partial \mathbf{R}_i} \right)_{\mathbf{R}=\mathbf{R}_0} \quad (15)$$

The transformation law under rotation follows straightforwardly from the chain rule: Write the rotated coordinates as

$$\mathbf{R}' = (\mathbf{I}_N \otimes \mathbf{Q}) \mathbf{R} \frac{\partial \mathbf{R}'}{\partial \mathbf{R}} = (\mathbf{I}_N \otimes \mathbf{Q}) \quad (16)$$

Then, define the energy in the rotated frame

$$E'(\mathbf{R}') = E((\mathbf{I}_N \otimes \mathbf{Q})^\top \mathbf{R}') = E(\mathbf{R}) \quad (17)$$

Then the gradient is

$$\nabla_{\mathbf{R}'} E'(\mathbf{R}') = (\mathbf{I}_N \otimes \mathbf{Q}) \nabla_{\mathbf{R}} E'(\mathbf{R}') \quad (18)$$

$$= (\mathbf{I}_N \otimes \mathbf{Q}) \nabla_{\mathbf{R}} E((\mathbf{I}_N \otimes \mathbf{Q})^{-1} (\mathbf{I}_N \otimes \mathbf{Q}) \mathbf{R}') \quad (19)$$

$$= (\mathbf{I}_N \otimes \mathbf{Q}) \nabla_{\mathbf{R}} E(\mathbf{R}) \quad (20)$$

showing the equivariance of the forces. The second-order derivative is

$$\mathbf{H}' = \nabla_{\mathbf{R}'}^2 E'(\mathbf{R}') = \nabla_{\mathbf{R}'} ((\mathbf{I}_N \otimes \mathbf{Q}) \nabla_{\mathbf{R}} E(\mathbf{R})) \quad (21)$$

$$= (\mathbf{I}_N \otimes \mathbf{Q}) \nabla_{\mathbf{R}'} \nabla_{\mathbf{R}} E(\mathbf{R}) \quad (22)$$

$$= (\mathbf{I}_N \otimes \mathbf{Q}) \nabla_{\mathbf{R}} \nabla_{\mathbf{R}} E(\mathbf{R}) (\mathbf{I}_N \otimes \mathbf{Q})^\top \quad (23)$$

$$= (\mathbf{I}_N \otimes \mathbf{Q}) \mathbf{H} (\mathbf{I}_N \otimes \mathbf{Q})^\top \quad (24)$$

which shows the Hessian’s equivariance.

4.1 Non-integrability of Hessians

Instead of relying on automatic differentiation, we propose to directly predict Hessians with an equivariant neural network. Similarly, direct force prediction has been proposed before [8, 16]. While direct force prediction introduces non-integrable errors (non-conservative fields), these errors only cause unbounded blowups when integrated over time, as in Molecular Dynamics (MD). In contrast, for geometry optimization and standard Hessian applications (ZPE, transition state search, preconditioning), these errors remain bounded and merely limit convergence precision rather than accumulating. Therefore, direct Hessian predictions are safe for mainstream computational chemistry tasks, analogous to widely used but non-integrable methods like BFGS.

Since direct Hessian predictions are not formulated as derivatives, they are not guaranteed to be integrable. It is important to understand when a non-integrability error can lead to catastrophically wrong downstream predictions similar to direct force predictions in MD simulations. Let’s review the more familiar case: Why do direct force predictions fail in MD simulations but succeed in geometry optimization? Suppose we directly predict forces with error $e(x) = F_{\text{pred}}(x) - F(x)$. We can decompose the error in a conservative and a non-conservative field using the Helmholtz decomposition: $e(x) = \nabla \phi(x) + S(x)$. $\phi \in \mathbb{R}$ is a scalar potential, and S is a vector field $S(x) \in \mathbb{R}^{3N}$ that is divergence-free $\nabla \cdot S(x) = 0$ but not curl-free $\exists x \text{ s.t. } \nabla \times S(x) \neq 0$. Now consider how the energy of the system changes over time in an MD simulation:

$$\begin{aligned} \Delta E' &= \int_0^T F_{\text{pred}}(x(t)) v(t) dt = \int_0^T (F(x(t)) + e(x(t))) v(t) dt \\ &= \int_0^T (F(x) + \nabla \phi(x) + S(x)) v(t) dt \\ &= E(T) - E(0) + \underbrace{\phi(T) - \phi(0)}_{\text{error bounded with time}} + \underbrace{\int_0^T S(x) v(t) dt}_{\text{error unbounded with time}} \end{aligned}$$

Since MD simulations use large T , non-conservative forces lead to the blowups in error.

In geometry optimization we define $E' = E + \phi$, then follow the gradient descent ODE:

$$\frac{dE'(x(t))}{dt} = \nabla E'(x(t)) \cdot \frac{dx(t)}{dt} \quad (25)$$

$$= -\nabla E'(x(t)) \cdot (\nabla E(x) + e(x)) \quad (26)$$

$$= -\nabla E'(x(t)) \cdot (\nabla E(x) + \nabla \phi(x) + S(x)) \quad (27)$$

$$= -\nabla E'(x(t)) \cdot (\nabla E'(x) + S(x)) \quad (28)$$

$$= -|\nabla E'(x(t))|^2 + \nabla E'(x) \cdot S(x) \quad (29)$$

Using the general inequality $2ab \leq a^2 + b^2$:

$$\frac{dE'(x(t))}{dt} = -|\nabla E'(x(t))|^2 + \nabla E'(x) \cdot S(x) \quad (30)$$

$$\leq -|\nabla E'(x(t))|^2 + \frac{1}{2}|\nabla E'(x(t))|^2 + \frac{1}{2}|S(x)|^2 \quad (31)$$

$$= -\frac{1}{2}|\nabla E'(x(t))|^2 + \frac{1}{2}|S(x)|^2 \quad (32)$$

Therefore, as long as $|\nabla E'(x(t))|^2 > |S(x)|^2$, the direct predicted force minimizes the energy plus some error potential similar to AD forces. Once we are close to convergence, we have $|\nabla E'(x(t))|^2 \leq |S(x)|^2$ and stop converging or enter a limit cycle. This means, we will converge to an extremum on an energy surface E' within a distance of $O(|S|)$. In practice, we are interested in how close we converge to the extrema of the DFT energy E . For this, it does not matter if the error stems from the error potential ϕ shifting the extrema, or S inducing a small limit cycle. As there is no time-dependent error blow-up, it is therefore safe to use direct forces for geometry optimization.

Now we can do the same for the directly predicted Hessian $H_{\text{pred}}(x) = H(x) + e(x)$: Denote the error again as $e(x) = \nabla^2 \phi(x) + S(x)$. As before, we would get errors scaling with integration length if we tried to calculate quantities of the form

$$Q(T) = \int_0^T f(x(t), H_{\text{pred}}(x(t))) dx(t) \quad (33)$$

leading to large errors for large T .

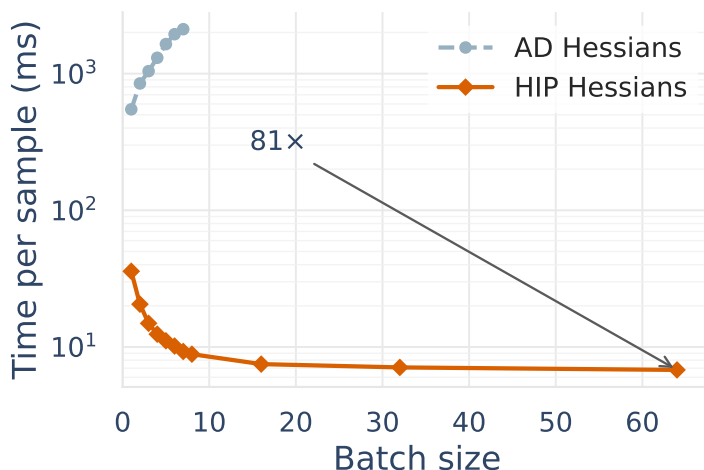
The key advantage is that none of the mainstream tasks integrate over a function of the Hessian for large T .

ZPE and geometry classification are single-point calculations; therefore, the errors $\nabla^2 \phi(x)$ and $S(x)$ behave similarly, and one error is not "worse" than the other. Mode following during the transition state search is equivalent to the geometry-optimization analysis with direct forces above. Energy minimization with the Hessian preconditioner is also unproblematic, as the preconditioner cannot change the minimum; it only changes the convergence speed.

In fact, it is common practice to use BFGS updates for transition state search and energy minimization, and BFGS updates do not have any notion of integrability either.

5 Batched AD Hessian performance

To test the performance of batched Hessians, we measure the inference time per processed sample for increasingly larger batch sizes and plot the results in Supplementary Figure 1. We observe rapid degradation in the AD Hessian speed with increasing batch size. To explain this, we have to understand exactly how AD treats Hessians. From the point of view of the AD engine, there is no difference between a batch of two



Supplementary Figure 1 Batch-wise Hessian timing. Mean wall-clock time per sample versus batch size for HIP Hessians (orange diamonds, solid) and AD Hessians (blue circles, dashed). AD timings stop at batch size 7 because larger batches exceeded available memory. HIP remains parallelizable to batch size 64. $n = 10$ repeats per method and batch size.

molecules of system size N_A and N_B , and a larger system of size $N_A + N_B$. In both cases, the MLIP function takes in an array of dimension $3N_A + 3N_B$, and returns either a single scalar $E_A + E_B$ or even worse, an array $[E_A, E_B]$, so mathematically the energy function looks like $E(\mathbf{R}) : \mathbb{R}^{3N_A+3N_B} \rightarrow \mathbb{R}$ or $E(\mathbf{R}) : \mathbb{R}^{3N_A+3N_B} \rightarrow \mathbb{R}^2$. Consequently, a Hessian of this function is a block diagonal matrix of dimension $3(N_A+N_B) \times 3(N_A+N_B)$, or even $2 \times 3(N_A+N_B) \times 3(N_A+N_B)$. Therefore, the memory costs grow quadratically with the batch size. Since we have to implement the Hessian calculation using sequential Hessian-Vector products to avoid running out of memory, we cannot even take advantage of parallelization in batched Hessian processing.

6 Additional results

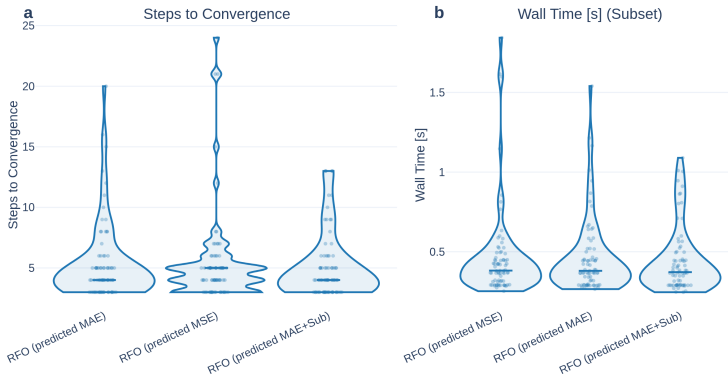
6.1 Loss function

A loss applied directly to eigenvectors requires an eigendecomposition of the predicted Hessian. Backpropagation through that decomposition is numerically unstable [47]. Gradients are finite only when $\mathbf{H}$ has distinct eigenvalues, and they become large when two eigenvalues approach each other because they depend on $\min_{i \neq j} 1/(\lambda_i - \lambda_j)$. We therefore use the subspace loss defined in the main text (Methods, Loss function design) rather than differentiating through the eigenspectrum.

We compare that subspace loss with mean squared error (MSE) and mean absolute error (MAE). On downstream tasks, subspace loss improves over MAE and MSE in (i) transition state search, as shown in Supplementary Table 3, and (ii) transition state identification by frequency analysis, as shown in Supplementary Table 4. We observe equal performance between MAE and the subspace loss for (i) geometry relaxation

Loss	Hessian ↓ eV/Å ²	Eigenvalues ↓ eV/Å ²	CosSim $\mathbf{v}_1$ ↑ unitless	λ_1 ↓ eV/Å ²	λ_2 ↓ eV/Å ²	λ_1^E ↓ eV/Å ²	λ_2^E ↓ eV/Å ²	CosSim $\mathbf{v}_1^E$ ↑ unitless	CosSim $\mathbf{v}_2^E$ ↑ unitless
MSE	0.033	0.072	0.728	0.168	0.088	0.037	0.013	0.959	0.928
MAE	0.026	0.057	0.825	0.127	0.052	0.031	0.012	0.976	0.952
MAE+Sub	0.030	0.063	0.870	0.130	0.030	0.031	0.010	0.980	0.957

Supplementary Table 1 Loss-function accuracy. Mean absolute errors and cosine similarities for HIP-EquiformerV2 trained with mean squared error (MSE), mean absolute error (MAE) or MAE plus subspace loss (MAE+Sub). Hessian, eigenvalue and λ_1 , λ_2 columns are in eV/Å². Superscript E denotes quantities after Eckart projection of the mass-weighted Hessian. Bold values mark the best result in each column. $n = 1000$ HORM-T1x validation geometries. No inferential statistical test was applied.



Supplementary Figure 2 Geometry relaxation by loss function. Rational-function optimization (RFO) with HIP-EquiformerV2 Hessians trained using mean squared error (MSE), mean absolute error (MAE) or MAE plus subspace loss (MAE+Sub). **(a)** Steps to convergence. **(b)** Wall-clock time. Violins show the sample distribution and overlaid points are individual geometries. $n = 80$ Transition1x validation reactant geometries. No inferential statistical test was applied.

(Supplementary Figure 2) and (ii) computing the zero-point energy (Supplementary Table 2).

Supplementary Table 1 shows that the subspace loss improves the first eigenvector $\mathbf{v}_1$ cosine similarity, and the second eigenvalue λ_2 MAE. This is consistent with the better performance of the subspace loss on transition state-related tasks, since the transition state is characterized via the two smallest eigenvalues λ_1, λ_2 , and is found by following the first eigenvector $\mathbf{v}_1$.

MSE generally underperforms compared to MAE and MAE with subspace loss. MSE has the theoretical advantage of being rotation-invariant, whereas MAE is not. In practice, however, we find that the MSE leads to a high variance (spikes) in the training loss and gradient norm across batches. We speculate that MAE outperforms MSE due to training stability. Hessian prediction with any loss function tested substantially reduces errors compared to AD Hessians.

6.2 Training resources

The Hessian dataset for Optimizing Reactive MLIP (HORM) provides the baseline checkpoints used here. Training AD Hessians is exceptionally more expensive than

Loss	Model	Reactant ZPE MAE (SD) [eV]	Δ ZPE MAE (SD) [eV]
MSE	HIP-EquiformerV2	0.0008 (0.0012)	0.0019 (0.0025)
MAE	HIP-EquiformerV2	0.0004 (0.0004)	0.0013 (0.0019)
MAE+Sub	HIP-EquiformerV2	0.0004 (0.0003)	0.0016 (0.0018)

Supplementary Table 2 Zero-point energy by loss function. Mean absolute error (MAE) of the reactant zero-point energy (ZPE) and of the reaction Δ ZPE relative to density-functional theory (DFT) for HIP-EquiformerV2 trained with mean squared error (MSE), mean absolute error (MAE) or MAE plus subspace loss (MAE+Sub). Values in parentheses are sample standard deviations of the signed errors $\text{ZPE}_{\text{model}} - \text{ZPE}_{\text{DFT}}$ and $\Delta\text{ZPE}_{\text{model}} - \Delta\text{ZPE}_{\text{DFT}}$. Bold values mark the best result in each column. Energies are in eV per molecule. $n = 47$ reactant-product pairs. No inferential statistical test was applied.

Loss	Model	TS Success	RFO Converged	Both
MSE	HIP-EquiformerV2	718	669	659
MAE	HIP-EquiformerV2	740	693	683
MAE+Sub	HIP-EquiformerV2	750	705	698

Supplementary Table 3 Transition-state search by loss function. Counts of ReactBench reactions that pass transition-state (TS) success, restricted-step partitioned rational-function optimization (RFO) convergence, or both, for HIP-EquiformerV2 trained with mean squared error (MSE), mean absolute error (MAE) or MAE plus subspace loss (MAE+Sub). Bold values mark the best result in each column. $n = 960$ reactions. No inferential statistical test was applied.

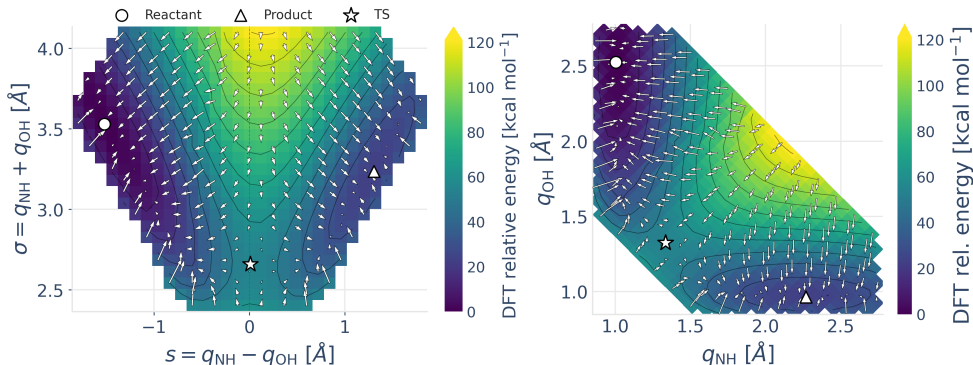
Name	TPR $\uparrow$	FPR $\downarrow$	FNR $\downarrow$	TNR $\uparrow$	Precision $\uparrow$	Accuracy $\uparrow$	Accuracy All $\uparrow$
MSE	88%	11%	12%	89%	86%	89%	86%
MAE	93%	7%	7%	93%	91%	93%	91%
MAE+Sub	93%	6%	7%	94%	92%	94%	92%

Supplementary Table 4 Frequency-analysis classification by loss function. True-positive rate (TPR), false-positive rate (FPR), false-negative rate (FNR), true-negative rate (TNR), precision and accuracy for identifying first-order transition states from the number of negative Hessian eigenvalues. Accuracy All is the accuracy for classifying any stationary-point type (minimum, first-order saddle or higher-order saddle). HIP-EquiformerV2 was trained with mean squared error (MSE), mean absolute error (MAE) or MAE plus subspace loss (MAE+Sub). Bold values mark the best result in each column. $n = 1000$ HORM-T1x validation geometries. No inferential statistical test was applied.

training HIP. The memory cost of computing loss gradients through the full $3N \times 3N$ Hessian is prohibitive. Instead, the baseline models from [30] randomly sample one or two columns of the Hessian via random Hessian-vector products at each training step. Despite this, training AD Hessians remains exceedingly expensive. The HORM paper states that they use 2 HVPs at a batch size of 128. Their checkpoint metadata shows a per-device batch size of 8, suggesting a $16 \times$ A30 GPU setup for 1 million gradient steps. In contrast, we trained HIP on a single H100 GPU for 500k steps. Additionally, each HIP training step costs 1/5 as much as AD Equiformer training on our hardware. Finally, we observe that HIP fully converges after 300-500k training steps, whereas AD Hessians require multiple steps due to the limited supervision and the inability to use the full Hessian.

Model	Hessian Columns	Batch Size	BZ / GPU	GPU	# GPUs	Training Steps
AlphaNet	1	32	16	H20	2	688,000
LeftNet-CF	1	64	16	A30	8	192,000
LeftNet-DF	2	64	16	A30	8	120,000
EquiformerV2 (E-F)	0	128	8	A30	16	384,000
EquiformerV2	2	128	8	A30	16	1,008,000
HIP-EquiformerV2	3N (all)	128	128	H100	1	500,000

Supplementary Table 5 Training resources. Hardware, Hessian-column sampling, batch size and training-step counts for HIP-EquiformerV2 compared with HORM checkpoints [30]. Hessian Columns is the number of Hessian-vector products used per training step; HIP uses the full $3N \times 3N$ Hessian.



Supplementary Figure 3 Glycine proton-transfer energy surface. Density-functional theory (DFT) relative energy at ω B97X-D3/6-31G(d) on the two-dimensional proton-transfer surface. Colour is DFT relative energy in kcal mol^{-1} . White arrows are forces in the plotted coordinates. The circle, star and triangle mark the DFT reactant, transition state and product, respectively. **(Left)** Coordinates $s = q_{\text{NH}} - q_{\text{OH}}$ and $\sigma = q_{\text{NH}} + q_{\text{OH}}$. **(Right)** Distances q_{NH} and q_{OH} . The DFT reactant and product have no imaginary frequencies; the DFT transition-state geometry has one. $n = 579$ constrained-relaxed geometries. No inferential statistical test was applied.

References

- [1] Thompson, A.P., Swiler, L.P., Trott, C.R., Foiles, S.M., Tucker, G.J.: Spectral neighbor analysis method for automated generation of quantum-accurate interatomic potentials. *Journal of Computational Physics* **285**, 316–330 (2015)
- [2] Shapeev, A.V.: Moment tensor potentials: A class of systematically improvable interatomic potentials. *Multiscale Modeling & Simulation* **14**(3), 1153–1173 (2016)
- [3] Bartók, A.P., Payne, M.C., Kondor, R., Csányi, G.: Gaussian approximation potentials: The accuracy of quantum mechanics, without the electrons. *Physical review letters* **104**(13), 136403 (2010)

- [4] Behler, J., Parrinello, M.: Generalized neural-network representation of high-dimensional potential-energy surfaces. *Physical review letters* **98**(14), 146401 (2007)
- [5] Schütt, K.T., Sauceda, H.E., Kindermans, P.-J., Tkatchenko, A., Müller, K.-R.: SchNet – A deep learning architecture for molecules and materials. *The Journal of Chemical Physics* **148**(24), 241722 (2018) <https://doi.org/10.1063/1.5019779>
- [6] Anderson, B., Hy, T.S., Kondor, R.: Cormorant: Covariant molecular neural networks. *Advances in neural information processing systems* **32** (2019)
- [7] Gasteiger, J., Groß, J., Günnemann, S.: Directional message passing for molecular graphs. In: *International Conference on Learning Representations* (2020). <https://openreview.net/forum?id=B1eWbxStPH>
- [8] Gasteiger, J., Becker, F., Günnemann, S.: GemNet: Universal Directional Graph Neural Networks for Molecules. In: *Advances in Neural Information Processing Systems*, vol. 34, pp. 6790–6802. Curran Associates, Inc. (2021)
- [9] Schütt, K., Unke, O., Gastegger, M.: Equivariant message passing for the prediction of tensorial properties and molecular spectra. In: *Proceedings of the 38th International Conference on Machine Learning*, pp. 9377–9388. PMLR (2021)
- [10] Batzner, S., Musaelian, A., Sun, L., Geiger, M., Mailoa, J.P., Kornbluth, M., Molinari, N., Smidt, T.E., Kozinsky, B.: E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. *Nature Communications* **13**(1), 2453 (2022) <https://doi.org/10.1038/s41467-022-29939-5>
- [11] Liu, Y., Wang, L., Liu, M., Lin, Y., Zhang, X., Oztekin, B., Ji, S.: Spherical message passing for 3d molecular graphs. In: *International Conference on Learning Representations* (2022). <https://openreview.net/forum?id=givsRXsOt9r>
- [12] Wang, Y., Wang, T., Li, S., He, X., Li, M., Wang, Z., Zheng, N., Shao, B., Liu, T.-Y.: Enhancing geometric representations for molecules with equivariant vector-scalar interactive message passing. *Nature Communications* **15**, 313 (2024) <https://doi.org/10.1038/s41467-023-43720-2>
- [13] Wang, Z., Liu, G., Zhou, Y., Wang, T., Shao, B.: Efficiently incorporating quintuple interactions into geometric deep learning force fields. *Advances in Neural Information Processing Systems* **36** (2023) <https://doi.org/10.52202/075280-3368>
- [14] Batatia, I., Kovacs, D.P., Simm, G., Ortner, C., Csanyi, G.: MACE: Higher Order Equivariant Message Passing Neural Networks for Fast and Accurate Force Fields. *Advances in Neural Information Processing Systems* **35**, 11423–11436 (2022)
- [15] Liao, Y.-L., Smidt, T.: Equiformer: Equivariant graph attention transformer for

- 3d atomistic graphs. In: International Conference on Learning Representations (2023). <https://openreview.net/forum?id=KwmPfARgOTD>
- [16] Liao, Y.-L., Wood, B.M., Das, A., Smidt, T.: Equiformerv2: Improved equivariant transformer for scaling to higher-degree representations. In: The Twelfth International Conference on Learning Representations (2024). <https://openreview.net/forum?id=mCOBKZmrzD>
 - [17] Burger, A., Thiede, L., Aspuru-Guzik, A., Vijaykumar, N.: DEQuify your force field: More efficient simulations using deep equilibrium models. In: AI for Accelerated Materials Design - ICLR 2025 (2025). <https://openreview.net/forum?id=XACVRYePQQ>
 - [18] Unke, O., Bogojeski, M., Gastegger, M., Geiger, M., Smidt, T., Müller, K.-R.: SE(3)-equivariant prediction of molecular wavefunctions and electronic densities. *Advances in Neural Information Processing Systems* **34**, 14434–14447 (2021)
 - [19] Yu, H., Xu, Z., Qian, X., Qian, X., Ji, S.: Efficient and equivariant graph networks for predicting quantum hamiltonian. In: International Conference on Machine Learning, pp. 40412–40424 (2023). PMLR
 - [20] Kaniselman, M., Miller, B.K., Gao, M., Nam, J., Levine, D.S.: Learning from the electronic structure of molecules across the periodic table. In: The Fourteenth International Conference on Learning Representations (2026). <https://openreview.net/forum?id=PS1YS8Wv4t>
 - [21] Li, Y., Xia, Z., Huang, L., Wei, X., Harshe, S., Yang, H., Luo, E., Wang, Z., Zhang, J., Liu, C., Shao, B., Gerstein, M.: Enhancing the scalability and applicability of kohn-sham hamiltonians for molecular systems. In: The Thirteenth International Conference on Learning Representations (2025). <https://openreview.net/forum?id=twEvvkQqPS>
 - [22] Febrer, P., Jørgensen, P.B., Pruneda, M., García, A., Ordejón, P., Bhowmik, A.: Graph2mat: universal graph to matrix conversion for electron density prediction. *Machine Learning: Science and Technology* **6**(2), 025013 (2025)
 - [23] Thiede, L.A., Aldossary, A., Burger, A., Campos-Gonzalez-Angulo, J.A., Zook, A., Alkan, M., Nakaji, K., Patti, T.L., Gonthier, J.F., Vakili, M.G., Aspuru-Guzik, A.: Coupled cluster con MōLe: Molecular orbital learning for neural wavefunctions. In: Proceedings of the 43rd International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 306. PMLR, Seoul, South Korea (2026). arXiv:2602.20232. <https://openreview.net/forum?id=i5QhkUCzJh>
 - [24] Burger, A., Thiede, L.A., Aldossary, A., Campos-Gonzalez-Angulo, J.A., Zook, A., Gonthier, J.F., Aspuru-Guzik, A.: MōLe-Λ: Learning the coupled-cluster response state for energies, gradients, and properties. In: AI4Physics: An ICML 2026 Workshop on AI for Physics, Seoul, South Korea (2026). arXiv:2605.29622.

<https://icml.cc/virtual/2026/72615>

- [25] Yang, X., Ma, H., Lu, Q., Bian, W.: Efficient Method for Numerical Calculations of Molecular Vibrational Frequencies by Exploiting Sparseness of Hessian Matrix. *The Journal of Physical Chemistry A* **128**(15), 3024–3032 (2024) <https://doi.org/10.1021/acs.jpca.3c07645>
- [26] Yuan, E.C.-Y., Kumar, A., Guan, X., Hermes, E.D., Rosen, A.S., Zádor, J., Head-Gordon, T., Blau, S.M.: Analytical ab initio Hessian from a deep learning potential for transition state optimization. *Nature Communications* **15**(1), 8865 (2024) <https://doi.org/10.1038/s41467-024-52481-5>
- [27] Gönnheimer, N., Reuter, K., Margraf, J.T.: Beyond Numerical Hessians: Higher-Order Derivatives for Machine Learning Interatomic Potentials via Automatic Differentiation. *Journal of Chemical Theory and Computation* **21**(9), 4742–4752 (2025) <https://doi.org/10.1021/acs.jctc.4c01790>
- [28] Rodriguez, A., Smith, J.S., Mendoza-Cortes, J.L.: Does Hessian Data Improve the Performance of Machine Learning Potentials? *Journal of Chemical Theory and Computation* (2025) <https://doi.org/10.1021/acs.jctc.5c00402>
- [29] Williams, N.J., Kabalan, L., Stojanovic, L., Zólyomi, V., Pyzer-Knapp, E.O.: Hessian QM9: A quantum chemistry database of molecular Hessians in implicit solvents. *Scientific Data* **12**(1), 9 (2025) <https://doi.org/10.1038/s41597-024-04361-2>
- [30] Cui, T., Han, Y., Jia, H., Duan, C., Zhao, Q.: A large scale molecular Hessian database for optimizing reactive machine learning interatomic potentials. *Scientific Data* **13**, 37 (2026) <https://doi.org/10.1038/s41597-025-06350-5>
- [31] Zhao, Q., Han, Y., Zhang, D., Wang, J., Zhong, P., Cui, T., Yin, B., Cao, Y., Jia, H., Duan, C.: Harnessing machine learning to enhance transition state search with interatomic potentials and generative models. *Advanced Science* **12**(34), 06240 (2025) <https://doi.org/10.1002/advs.202506240>
- [32] Christensen, A.S., Von Lilienfeld, O.A.: On the role of gradients for machine learning of molecular energies and forces. *Machine Learning: Science and Technology* **1**(4), 045018 (2020) <https://doi.org/10.1088/2632-2153/abba6f>
- [33] Fu, X., Wood, B.M., Barroso-Luque, L., Levine, D.S., Gao, M., Dzamba, M., Zitnick, C.L.: Learning smooth and expressive interatomic potentials for physical property prediction. In: *Proceedings of the 42nd International Conference on Machine Learning. Proceedings of Machine Learning Research*, vol. 267, pp. 17875–17893. PMLR, Vancouver, Canada (2025). <https://proceedings.mlr.press/v267/fu25h.html>
- [34] Wander, B., Musielewicz, J., Cheula, R., Kitchin, J.R.: Accessing numerical

- energy Hessians with graph neural network potentials and their application in heterogeneous catalysis. *The Journal of Physical Chemistry C* **129**(7), 3510–3521 (2025)
- [35] Christiansen, M.-P.V., Rønne, N., Hammer, B.: Atomistic global optimization X: A Python package for optimization of atomistic structures. *The Journal of Chemical Physics* **157**(5), 054701 (2022) <https://doi.org/10.1063/5.0094165>
 - [36] Rønne, N., Christiansen, M.-P.V., Slavensky, A.M., Tang, Z., Brix, F., Pedersen, M.E., Bisbo, M.K., Hammer, B.: Atomistic structure search using local surrogate model. *The Journal of Chemical Physics* **157**(17), 174115 (2022) <https://doi.org/10.1063/5.0121748>
 - [37] Westermayr, J., Gilkes, J., Barrett, R., Maurer, R.J.: High-throughput property-driven generative design of functional organic molecules. *Nature Computational Science* **3**(2), 139–148 (2023) <https://doi.org/10.1038/s43588-022-00391-1>
 - [38] Rønne, N.: Generative diffusion model for surface structure discovery. *Physical Review B* **110**(23) (2024) <https://doi.org/10.1103/PhysRevB.110.235427>
 - [39] Bitzek, E., Koskinen, P., Gähler, F., Moseler, M., Gumbsch, P.: Structural Relaxation Made Simple. *Physical Review Letters* **97**(17), 170201 (2006) <https://doi.org/10.1103/PhysRevLett.97.170201>
 - [40] Simons, J., Joergensen, P., Taylor, H., Ozment, J.: Walking on potential energy surfaces. *The Journal of Physical Chemistry* **87**(15), 2745–2753 (1983) <https://doi.org/10.1021/j100238a013> . Accessed 2025-09-16
 - [41] Banerjee, A., Adams, N., Simons, J., Shepard, R.: Search for stationary points on surfaces. *The Journal of Physical Chemistry* **89**(1), 52–57 (1985) <https://doi.org/10.1021/j100247a015>
 - [42] Henkelman, G., Jónsson, H.: A dimer method for finding saddle points on high dimensional potential surfaces using only first derivatives. *The Journal of Chemical Physics* **111**(15), 7010–7022 (1999) <https://doi.org/10.1063/1.480097>
 - [43] Henkelman, G., Uberuaga, B.P., Jónsson, H.: A climbing image nudged elastic band method for finding saddle points and minimum energy paths. *The Journal of Chemical Physics* **113**(22), 9901–9904 (2000) <https://doi.org/10.1063/1.1329672> . Accessed 2025-09-11
 - [44] Peters, B., Heyden, A., Bell, A.T., Chakraborty, A.: A growing string method for determining transition states: Comparison to the nudged elastic band and string methods. *The Journal of Chemical Physics* **120**(17), 7877–7886 (2004) <https://doi.org/10.1063/1.1691018> . Accessed 2025-09-11
 - [45] Zimmerman, P.M.: Single-ended transition state finding with the growing string

- method. *Journal of Computational Chemistry* **36**(9), 601–611 (2015) <https://doi.org/10.1002/jcc.23833>
- [46] Louck, J.D., Galbraith, H.W.: Eckart vectors, Eckart frames, and polyatomic molecules. *Reviews of Modern Physics* **48**(1), 69–106 (1976) <https://doi.org/10.1103/RevModPhys.48.69>
- [47] Aa, N., Morsche, H., Mattheij, R.: Computation of eigenvalue and eigenvector derivatives for a general complex-valued eigensystem. *The Electronic Journal of Linear Algebra* **16**, 300–314 (2007) <https://doi.org/10.13001/1081-3810.1203>